

DengarTrack

Digital Newborn Hearing Screening Management Platform

User Guide & Demo Briefing

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Scan QR codes on the last page to access the app and web dashboard

1. What is DengarTrack?

DengarTrack is a digital platform designed to replace paper-based newborn hearing screening workflows in Malaysian public hospitals. It was built for Siti Aminah Kamaludin's PhD research on reducing Loss-to-Follow-Up (LTFU) rates from 29% to under 10%.

The platform connects four types of hospital users through one unified system: ward screeners, hospital coordinators (audiologists), UNHS national coordinators, and Ministry of Health officers. All data is encrypted, PDPA 2010 compliant, and stored in Malaysia.

Key Features

- Point-of-care screening entry -- QR wristband scan, ear result buttons, under 60 seconds
- Offline-first mobile app -- works without WiFi, auto-syncs when connected
- Real-time coordinator dashboard -- follow-up queue, coverage rates, 1-3-6 KKM benchmarks
- Automated MoH reporting -- one-click Excel export in KKM format
- Full audit trail -- every action timestamped, immutable, PDPA compliant
- Bilingual -- Bahasa Melayu default, English toggle on every screen

2. How to Access DengarTrack

There are two ways to use DengarTrack: the **mobile app** for clinical staff at the ward, and the **web dashboard** for coordinators and administrators.

Mobile App (DengarTrack)

Designed for Android phones and tablets (Android 9+). Screeners and coordinators use this for screening entry, follow-up actions, and shift summaries. Scan the QR code on the last page to download the APK directly to your Android device.

Note: iOS is not supported for this demo. The app is Android-only.

Web Dashboard

Designed for desktop and laptop browsers. Coordinators, UNHS coordinators, and MoH officers use this for programme overview, user management, and national reporting. Scan the QR code on the last page or visit the URL directly in any browser.

Recommended browsers: Chrome, Edge, Firefox. Safari is acceptable but not optimised.

3. Demo Credentials -- All Accounts

PIN for **ALL**demo users: **1234**

Use these accounts to test every role in the system. Each role sees different data and screens.

Role	Staff ID	Hospital	What They Do
Screener	SCR001HKL	HKL001	Records screenings at cot side
Screener	SCR001HPJ	HPJ001	Records screenings at cot side
Screener	SCR001HSB	HSB001	Records screenings at cot side
Coordinator	COO001HKL	HKL001	Manages hospital programme
Coordinator	COO001HPJ	HPJ001	Manages hospital programme
Coordinator	COO001HSB	HSB001	Manages hospital programme
UNHS Coordinator	ADM001	All hospitals	National oversight, audit logs
MoH / KKM	MOH001	National view	Aggregate reporting only

For roles **SCR** & **COO**, there are a total of **3 Users per hospital**. Just change the last digit to 2,3.

Hospitals in demo: **HKL001** (Hospital KL), **HPJ001** (Hospital Putrajaya), **HSB001** (Hospital Sungai Buloh)
Babies seeded: **60 total** across all hospitals For Demo purposes 5 babies for each hospitals in included in the wristbandQR PDF

4. Screener Role -- Mobile App Walkthrough

The screener is the nurse or audiologist performing OAE/AABR screening at the baby's cot. This is **the most frequent user of the system**.

Step 1: Login

1. Open the DengarTrack app
2. Select your hospital from the dropdown
3. Enter Staff ID (e.g., SCR001HKL)
4. Enter PIN: **1234**
5. Tap **Log Masuk / Sign In**
6. Toggle BM/EN at top right if needed

Step 2: Home Screen

After login, you see:

- Your name and role badge (green = Screener)
- Sync status (green = all saved, yellow = pending sync)
- Two big buttons: **Saringan Baharu / New Screening** and **Ringkasan Syif / Shift Summary**

Step 3: New Screening

1. Tap **Saringan Baharu**
2. Scan QR on baby's wristband OR tap **Enter Manually** and type baby ID
3. Baby info appears (ID, Ward) -- confirm it is correct
4. Select screening type: **TEOAE, AABR, or ABR**
5. Tap **LULUS / PASS** (green) or **RUJUK / REFER** (red) for each ear
6. Add optional notes (max 100 characters)
7. Tap **Hantar / Submit**
8. Green checkmark = saved. If offline, it queues for auto-sync.

Important: If either ear is RUJUK, the system automatically creates a follow-up task for the coordinator. No extra action needed from the screener.

Step 4: Shift Summary

Tap **Ringkasan Syif** to see:

- Total screened today, passed, referred
- Motivational message
- Toggle **Today / All History** to see past screenings
- Baby IDs are masked (****Y004) for privacy
- Green border = PASS, red border = REFER

5. Coordinator Role -- Mobile + Web

The coordinator (audiologist) manages the entire hospital's UNHS programme. They can use both mobile app and web dashboard.

Mobile App -- Coordinator Dashboard

Login with COO001HKL (or HPJ/HSB), PIN 1234.

The dashboard shows:

- **KPI Cards:** Total screenings, PASS count, REFER count, Follow-up queue, LTFU count
- **Last Screening Time:** When the most recent screening was done
- **Active Screeners:** How many screeners worked today
- **Coverage Rate:** Circular progress -- screened babies vs registered babies
- **1-3-6 KKM Benchmark:** Two progress bars (screened by 1 month, diagnosed by 3 months)
- **Ward Breakdown:** Table showing screenings per ward with refer rates
- **Follow-up Queue:** All pending refer cases with action buttons
- **Today's Screenings:** Toggle to All History to see everything

Follow-up Actions

For each refer case, tap to see details and choose:

- **Tandakan Dihubungi / Mark Contacted** -- you called the parent
- **Buat Temujanji / Book Appointment** -- scheduled follow-up date • **Eskalasi / Escalate** -- needs senior attention
- **Selesai / Complete** -- baby attended follow-up
- **Tanda LTFU / Mark LTFU** -- lost to follow-up after 60 days

Every action is logged in the audit trail with your name and timestamp.

Web Dashboard -- Coordinator Portal

Login at the web URL with same credentials. The web view is richer for desk work:

- **Dashboard (/coordinator):** Same KPIs as mobile but with charts and trend lines
- **Follow-ups (/coordinator/followups):** Full follow-up queue with search, filters, bulk actions
- **Babies (/coordinator/babies):** Register new babies -- system auto-generates HLX ID
- **Users (/coordinator/users):** Manage screeners -- create, deactivate, reset PIN
- **Reports (/coordinator/reports):** Monthly report view + one-click Excel export
- **Screenings (/coordinator/screenings):** Full history of all hospital screenings

6. UNHS Coordinator & MoH Roles -- Web Only

UNHS Coordinator (ADM001)

Login with ADM001, PIN 1234. No hospital needed -- this is a national oversight role.

- **Dashboard (/unhs):** Aggregate monthly metrics across all hospitals, recent audit activity log
- **Users (/unhs/users):** Create and manage hospital coordinators (COO001HKL, etc.)
- **Cannot see:** Individual baby records, specific screening details, or personal identifiers
- **Can see:** Summary counts, trends, audit logs of who did what and when

MoH / KKM Officer (MOH001)

Login with MOH001, PIN 1234. The most restricted view by design.

- **Dashboard (/moh):** National aggregate only -- total hospitals, total screenings, coverage rates
- **Hospital Performance Table:** Each hospital's coverage rate, refer rate, LTFU rate
- **Cannot see:** Any individual baby, any screener name, any hospital coordinator detail
- **Can see:** Programme-level metrics for policy and resource planning

This is intentional. MoH does not need individual records for national reporting. The system enforces this at the API level -- it cannot be bypassed.

7. Recommended Demo Flow for i-ACHIEVE 2026

Story 1: The Problem (1 minute)

- 29% LTFU rate in Malaysia vs 10% international benchmark
- Paper cards get lost, Excel is manual, WhatsApp violates PDPA
- Coordinators spend 2-4 hours monthly on MoH reports
- **DengarTrack fixes all of this in one platform**

Story 2: The Screener (2 minutes)

1. Show login screen -- select hospital, enter SCR001HKL, PIN 1234
2. Show home screen -- sync status, role badge
3. Tap New Screening -- scan QR or manual entry
4. Show baby info card -- confirm correct baby
5. Tap PASS for left ear, REFER for right ear
6. Submit -- green confirmation
7. Show Shift Summary -- today's work, toggle to All History
8. **Key message:** Under 60 seconds per entry, works offline, no training needed

Story 3: The Coordinator (2 minutes)

1. Switch to web dashboard -- login COO001HKL
2. Show KPI cards -- numbers updated in real-time from screener entry
3. Show Follow-up Queue -- the REFER case just appeared automatically
4. Click into follow-up -- show status dropdown, appointment date, notes
5. Show 1-3-6 Benchmark -- progress bars, KKM target $\geq 90\%$
6. Show Reports -- monthly totals, click Export Excel
7. **Key message:** No more manual reports, no more lost cases

Story 4: National Oversight (1 minute)

1. Login as ADM001 (UNHS) -- show aggregate dashboard, audit logs
2. Login as MOH001 -- show national summary, hospital performance table
3. **Key message:** Data-driven policy, no individual privacy risk, full accountability

8. Troubleshooting & Quick Fixes

App won't download from QR?

- Android may block unknown sources -- go to Settings > Security > Allow Unknown Sources. Try Chrome browser instead of default browser. If still failing, use the direct URL instead of QR.

Login says Invalid credentials?

- Check Staff ID spelling -- it is case-sensitive (SCR001HKL not scr001hkl). PIN is 1234 for all demo accounts. Make sure you selected the correct hospital. If account was deactivated, use a different one.

Dashboard shows empty or No data?

- This is normal for fresh logins -- data appears after first screening entry. Try the screener flow first, then check coordinator dashboard. If still empty after 30 seconds, refresh the page.

Web dashboard won't load?

- Check internet connection -- the web needs connectivity (unlike mobile offline mode). Try a different browser. The URL is on Vercel -- if it is down, the mobile app still works with Railway backend.

BM/EN toggle not working?

- Toggle is at top-right on every screen. Web toggle is in the top bar. Language preference saves automatically -- no need to reselect each time.

9. Technical Overview

Backend: FastAPI (Python) + PostgreSQL, deployed on Railway

Mobile: Flutter (Dart), Android 9+, offline SQLite with auto-sync and Shorebird for OTA updates.

Web: Next.js 16 + React 19 + Tailwind CSS, deployed on Vercel

Auth: JWT tokens + 4-role RBAC enforced at API level

Security: TLS 1.3 in transit, AES-256 at rest, PDPA 2010 compliant

Data residency: All servers in Malaysia region

Integration ready: HL7 FHIR R4 for future KKM registry, MySejahtera, HIS/iPerak

10. QR Codes -- Scan or Tap to Access

Left: Download DengarTrack Mobile App (Android APK)

Right: Open Web Dashboard in Browser

You can scan the QR with any camera, or tap/click the QR image directly to open the link. Print this page for the booth. Have one Android tablet ready as backup.

	
DengarTrack App	DengarTrack Web Dashboard
Tap or Scan DengarTrack App	Tap or Scan DengarTrack Web Dashboard

Direct URLs (if QR fails):

Mobile App: <https://github.com/UwaisZulkarnain/HearlinX/releases/latest/download/DengarTrack.apk>

Web Dashboard: <https://dengartrackwebdashboard.vercel.app>

For technical support during demo, contact Uwais (developer) or check the GitHub repository.